

Christian Garry

MSc Scientific Computing | Probability · Statistics · Optimisation | C++/Python
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Education

MSc Scientific Computing & Data Analysis (AI for Engineering) Durham University	Durham, United Kingdom Sep 2025 – Present
• Focus: Probability and statistics, optimisation, numerical methods, Bayesian and regression models, HPC/GPU programming.	
MEng Electronic Engineering Durham University	Durham, United Kingdom Sep 2020 – Jun 2024

Experience

Industrial Tutor Durham University, Department of Engineering	Durham, United Kingdom Sep 2025 – Present
• Lead weekly design tutorials; mentor teams on scope, feasibility, Gantt planning, and design reviews. • Coach teams through feasibility-driven decision-making: surface assumptions and specifications, test them, and retire designs that fail cost, risk, or performance constraints.	
Graduate Communications Engineer Siemens PLC	Hebburn, United Kingdom Sep 2024 – Present
• Implemented and adapted C/C++ communication components (TCP/IP, serial) to support data exchange between simulated systems and PC-based engineering tools. • Built Python-based engineering tooling to structure and query internal technical documentation and codebases using a retrieval-augmented workflow. • Integrated local LLMs (via Ollama) with a vector database (Qdrant) for context-aware document retrieval, focusing on traceability and reproducible behaviour rather than model optimisation. • Adapted existing network signal and waveform-generation utilities to interface with specific IP endpoints for development and testing workflows. • Supported validation and testing activities by exercising communication paths and identifying failure modes during integration.	

Projects & Research

Silicon Carbide JFET CPU Master's Dissertation	Durham, United Kingdom Oct 2023 – Apr 2024
• Independently designed and simulated a 4-bit CPU using SiC JFET logic in LTspice, targeting computation in extreme temperature and radiation environments. • Constructed a full verification toolchain (C++ compiler, assembler, Python automation) and validated instruction-level behaviour via emulator parity checks and waveform-based state reconstruction. • Investigated architectural trade-offs, race conditions, and storage constraints, explicitly documenting assumptions, limitations, and scalability bottlenecks.	

Key Skills

Probability & Statistics; Numerical Methods; Optimisation; Monte Carlo Simulation; Scientific Computing; Python; C++; High-Performance Computing

Leadership, Activities & Interests

Leadership: Bishops Diocesan College Fencing Team Captain; Durham Fresher Representative (2022–23).
Activities: Durham University Fencing Team; Boxing (Student Fight Night).
Achievements: Weldon le Huray Fencing Scholarship (2020–24); South Africa U17 Fencing Champion; President's Award (Gold).